

**1. Write a Pandas program to create a scatter plot of the trading volume/stock prices of Alphabet Inc. stock between two specific dates.**

**alphabet\_stock\_data:**

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Alphabet stock data
data = {
    'Date': ['2023-01-02', '2023-01-10', '2023-01-18',
            '2023-02-05', '2023-02-15', '2023-03-01',
            '2023-03-10', '2023-03-20', '2023-04-05',
            '2023-04-15'],
    'Close': [91.80, 94.50, 96.20, 98.75, 101.30,
             103.60, 105.40, 107.80, 110.20, 112.50],
    'Volume': [25000000, 28000000, 26000000, 30000000,
              31000000, 27000000, 29000000, 32000000,
              33000000, 35000000]
}

# Create DataFrame
df = pd.DataFrame(data)

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Select data between two dates
start_date = '2023-01-01'
```

```
end_date = '2023-03-31'
```

```
filtered_df = df[(df['Date'] >= start_date) &  
                  (df['Date'] <= end_date)]
```

```
# Create Scatter Plot
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(filtered_df['Volume'],  
            filtered_df['Close'])
```

```
plt.title('Alphabet Inc. Stock: Trading Volume vs Stock Price')
```

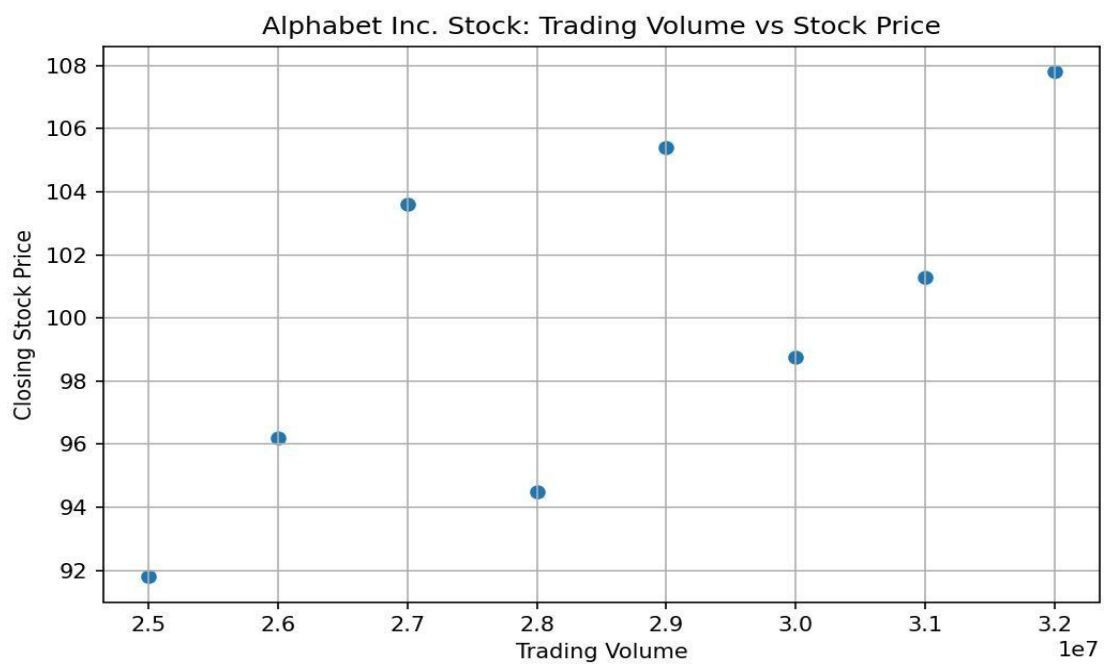
```
plt.xlabel('Trading Volume')
```

```
plt.ylabel('Closing Stock Price')
```

```
plt.grid(True)
```

```
plt.show()
```

Output:



**2. Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.(refer sales\_data table)**

Code:

```
import pandas as pd

data = {
    'Item': ['Television', 'Home Theater', 'Television', 'Cell Phone',
            'Television', 'Home Theater', 'Television', 'Television',
            'Television', 'Home Theater', 'Television', 'Home Theater',
            'Home Theater', 'Television', 'Desk', 'Video Games',
            'Home Theater', 'Cell Phone'],

    'Sale_amt': [113810, 25000, 43128, 6075, 67088, 30000,
                89850, 107820, 38336, 30000, 107820,
                14500, 40500, 41930, 250, 936,
                14000, 14400]
}

df = pd.DataFrame(data)

# Pivot Table
pivot_table = pd.pivot_table(
    df,
    values='Sale_amt',
    index='Item',
    aggfunc=['max', 'min']
)

print(pivot_table)
```

Output:

|              | max      | min      |
|--------------|----------|----------|
| Item         | Sale_amt | Sale_amt |
| Cell Phone   | 14400    | 6075     |
| Desk         | 250      | 250      |
| Home Theater | 40500    | 14000    |
| Television   | 113810   | 38336    |
| Video Games  | 936      | 936      |